Case Study 6: Student Performance Trend Analysis

1. Problem Statement

Compare scores from three assessments, calculate changes, identify the lowest-scoring subject, and classify the student using the average score.

2. Objectives

Store marks in lists.

Calculate averages and score changes.

Use conditions to classify performance.

3. Assumptions

The student has three subjects and two assessments.

Each mark is between 0 and 100.

Average 75 or above is Excellent; 50 to below 75 is Good; below 50 is Needs Improvement.

4. Algorithm

1. Start.

2. Input three subject names.

3. Input the first and second assessment marks for each subject.

4. Calculate each subject's change and the overall average of second-assessment marks.

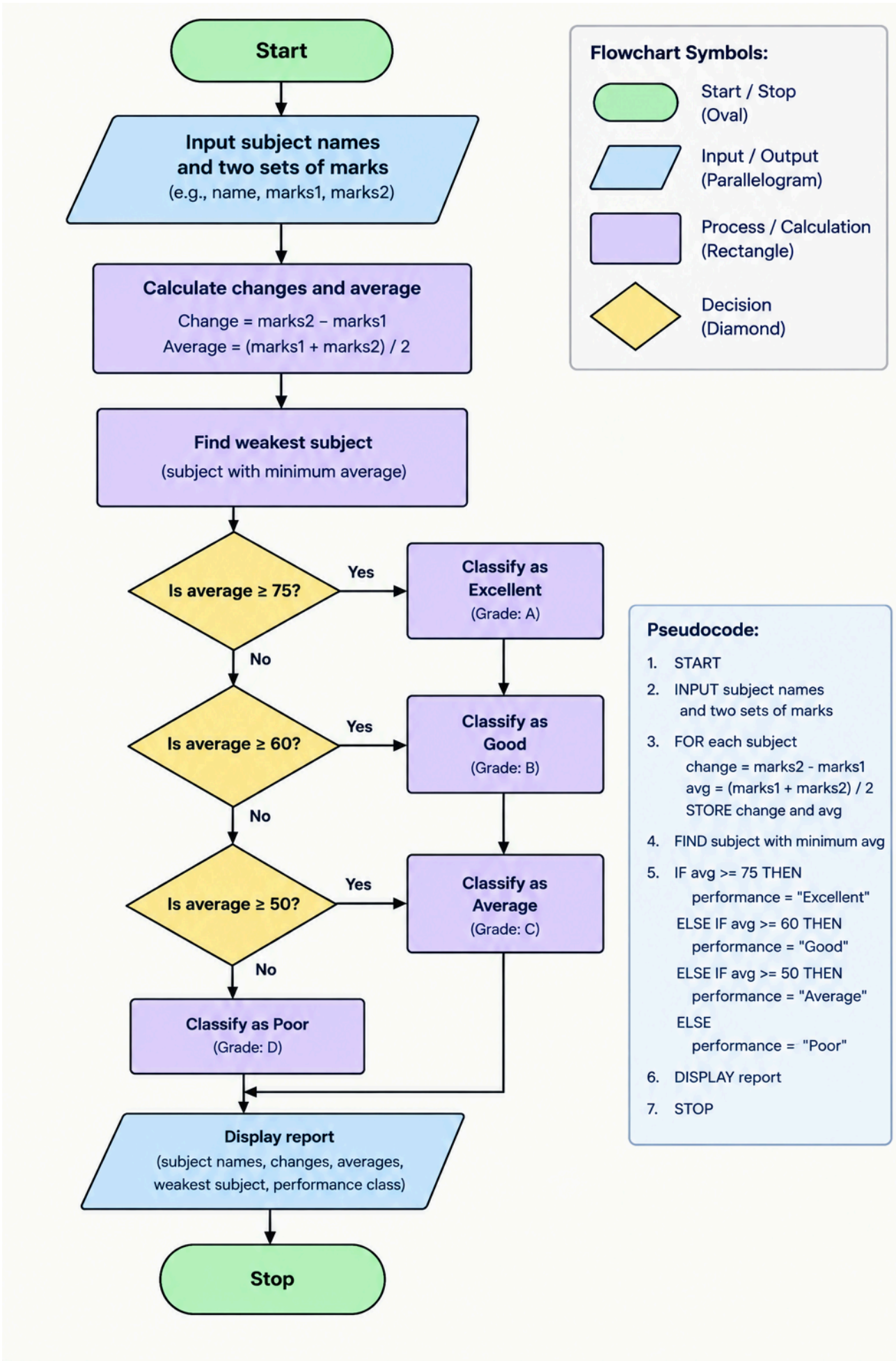
5. Find the subject with the lowest second-assessment mark.

6. Classify performance based on the average.

7. Display the report.

8. Stop.

5. Flowchart (text form)


Start
Input subject names and two sets of marks
(e.g., name, marks1, marks2)
Calculate changes and average
Change = marks2 – marks1
Average = (marks1 + marks2) / 2
Find weakest subject
(subject with minimum average)
Is average ≥ 75?
Yes
Classify as Excellent
(Grade: A)
No
Is average ≥ 60?
Yes
Classify as Good
(Grade: B)
No
Is average ≥ 50?
Yes
Classify as Average
(Grade: C)
No
Classify as Poor
(Grade: D)
Display report
(subject names, changes, averages, weakest subject, performance class)
Stop
Flowchart Symbols:
Start / Stop (Oval)
Input / Output (Parallelogram)
Process / Calculation (Rectangle)
Decision (Diamond)
Pseudocode:
1. START
2. INPUT subject names and two sets of marks
3. FOR each subject
change = marks2 - marks1
avg = (marks1 + marks2) / 2
STORE change and avg
4. FIND subject with minimum avg
5. IF avg >= 75 THEN
performance = "Excellent"
ELSE IF avg >= 60 THEN
performance = "Good"
ELSE IF avg >= 50 THEN
performance = "Average"
ELSE
performance = "Poor"
6. DISPLAY report
7. STOP

6. Pseudocode

```
START
INPUT 3 subject names, first marks and second marks
FOR each subject: change = second mark - first mark
average = sum(second marks) / 3
Find subject with minimum second mark
IF average >= 75: Excellent; ELSE IF average >= 50: Good; ELSE Needs
Improvement DISPLAY changes, average, weakest subject, classification
STOP
```

7. Complete Python Source Code

```
subjects = []
first_marks = []
second_marks = []
```



```
for i in range(3):
```

subject = input(f"Enter subject {i + 1} name: ")

first = float(input("First assessment mark (0-100): "))

second = float(input("Second assessment mark (0-100): "))

```
  while first < 0 or first > 100 or second < 0 or second > 100:
  print("Marks must be between 0 and 100.")
```

first = float(input("Re-enter first assessment mark: "))

second = float(input("Re-enter second assessment mark: "))

subjects.append(subject)

first_marks.append(first)

second_marks.append(second)

changes = []

```
for i in range(3):
```

changes.append(second_marks[i] - first_marks[i])

```
average = sum(second_marks) / len(second_marks)
```

weakest_index = second_marks.index(min(second_marks))

```
if average >= 75:
```

category = "Excellent"

elif average >= 50:

category = "Good"

```
else:
```

category = "Needs Improvement"

```
print("
----- PERFORMANCE REPORT -----")
for i in range(3):
 print(f"{subjects[i]}: change = {changes[i]:+.1f} marks")
print(f"Second-assessment average: {average:.2f}")
print("Lowest-scoring subject:", subjects[weakest_index])
print("Performance category:", category)
```

8. Test Case and Expected Output

Maths: 60→70; English: 80→75; Python: 45→55.

Maths: change = +10.0 marks English: change = -5.0 marks Python: change = +10.0 marks
Second-assessment average: 66.67 Lowest-scoring subject: Python Performance category: Good

9. Conclusion

The program analyses assessment changes, identifies the lowest-scoring subject, and classifies overall performance.